EXPERIMENT 7

Producer–Consumer Problem using Semaphores

```
#include <stdio.h>
#include <pthread.h>
#include <semaphore.h>
#include <unistd.h>

#define SIZE 5

int buffer[SIZE];
int in = 0, out = 0;

sem_t empty, full, mutex;

void* producer(void* p){
    int item, i;
    for(i = 0; i < 5; i++){
        item = i + 1;

        sem_wait(&empty);
        sem_wait(&mutex);

        buffer[in] = item;
        printf("Produced: %d at position %d\n", item, in);
        in = (in + 1) % SIZE;
```

```
        sem_post(&mutex);
        sem_post(&full);

        sleep(1);
    }
    return NULL;
}

void* consumer(void* p){
    int item, i;
    for(i = 0; i < 5; i++){

        sem_wait(&full);
        sem_wait(&mutex);

        item = buffer[out];
        printf("Consumed: %d from position %d\n", item, out);
        out = (out + 1) % SIZE;

        sem_post(&mutex);
        sem_post(&empty);

        sleep(1);
    }
    return NULL;
}

int main(){
    pthread_t p, c;
```

```
    sem_init(&empty, 0, SIZE);
    sem_init(&full, 0, 0);
    sem_init(&mutex, 0, 1);

    pthread_create(&p, NULL, producer, NULL);
    pthread_create(&c, NULL, consumer, NULL);

    pthread_join(p, NULL);
    pthread_join(c, NULL);

    sem_destroy(&empty);
    sem_destroy(&full);
    sem_destroy(&mutex);

    return 0;
}
```

OUTPUT

```
Produced: 1 at position 0
Consumed: 1 from position 0
Produced: 2 at position 1
Consumed: 2 from position 1
Produced: 3 at position 2
Consumed: 3 from position 2
Produced: 4 at position 3
Consumed: 4 from position 3
Produced: 5 at position 4
Consumed: 5 from position 4
```